

A Bioinspired Distributed Optical Fiber Sensing Device for Wearable Thermomechanical Monitoring of Breast Tissue

Vincenzo Lavorgna

e-mail address: vincenzo.lavorgna@unicampus.it

Unit of Measurement and Biomedical Instrumentation

Departmental Faculty of Engineering

Università Campus Bio-Medico di Roma, Rome 00128, Italy

Marco Gallinaro

e-mail address: m.gallinaro@unicampus.it

Unit of Measurement and Biomedical Instrumentation

Departmental Faculty of Engineering

Università Campus Bio-Medico di Roma, Rome 00128, Italy

Andrea Polimadei

e-mail address: andrea.polimadei@enea.it

Photonics Micro- and Nanostructures Laboratory

ENEA Research Center of Frascati, Frascati 00044, Italy

Rosaria D'Amato

e-mail address: rosaria.damato@enea.it

Photonics Micro- and Nanostructures Laboratory

ENEA Research Center of Frascati, Frascati 00044, Italy

Michele A. Caponero

e-mail address: michele.caponero@enea.it

Photonics Micro- and Nanostructures Laboratory

ENEA Research Center of Frascati, Frascati 00044, Italy

Emiliano Schena*

e-mail address: e.schena@unicampus.it

Unit of Measurement and Biomedical Instrumentation

Departmental Faculty of Engineering

Università Campus Bio-Medico di Roma, Rome 00128, Italy

Fondazione Policlinico Universitario Campus Bio-Medico di Roma, Rome 00128, Italy

Daniela Lo Presti

e-mail address: d.lopresti@unicampus.it

Unit of Measurement and Biomedical Instrumentation

Departmental Faculty of Engineering

Università Campus Bio-Medico di Roma, Rome 00128, Italy

Fondazione Policlinico Universitario Campus Bio-Medico di Roma, Rome 00128, Italy

* corresponding author

Keywords: distributed optical fiber sensing, wearable device, breast tissue monitoring, thermomechanical sensing, bioinspired design, optical frequency-domain reflectometry, breast cancer

Highlights

- A wearable distributed optical fiber sensing platform was bioinspired by orb-weaving spider webs for thermomechanical mapping and monitoring of breast tissue.
- The single-fiber spiral layout translates both web-like area coverage and continuous-filament perturbation localization, providing 458 sensing sites over an annular surface.
- Static mechanical and thermal characterization protocols were performed under conditions representative of the intended wearable application, enabling the estimation of local force and temperature sensitivities.
- Mechanical phantom tests demonstrated the ability of the device to qualitatively detect and spatially remap localized stiffness alterations induced by a rigid embedded inclusion.
- Thermal phantom tests showed the capability of the device to qualitatively identify and spatially remap localized temperature perturbations.

ABSTRACT

Breast tissue assessment relies on established imaging techniques essential for diagnosis but less suitable for frequent wearable-based monitoring. Sensing physical parameters, such as stiffness- and temperature-related variations, may provide complementary information on breast tissue conditions. Here, we present a soft, bioinspired wearable platform based on distributed optical fiber sensing for distributed thermomechanical monitoring at the breast surface. The system consists of a thin annular silicone patch embedding a single optical fiber arranged as an Archimedean spiral, inspired by the

capture spiral of orb-weaving spider webs. This layout translates the functional concept of an extended filamentous structure acting as a distributed mechanosensory medium for spatial localization of perturbations. Interrogated through optical frequency-domain reflectometry, the embedded fiber provides 458 sensing sites remapped over the annular active region, without arrays of discrete sensors. The platform is conceived for future integration into wearable supports, where gentle contact with the breast surface could transfer stiffness- or temperature-related alterations to the fiber as spatially resolved spectral-shift variations. It was characterized under controlled compression and temperature variations, yielding mean force and temperature sensitivities of -0.71 GHz/N and $-2.25 \text{ GHz/}^{\circ}\text{C}$, respectively. Preliminary tests were then performed on breast-mimicking phantoms designed to reproduce two classes of localized alterations, namely a rigid mechanical inhomogeneity and a localized thermal perturbation. In both cases, the proposed system enabled qualitative identification and spatial remapping of the imposed alterations. These results establish the feasibility of a fully distributed, single-fiber platform as a first step toward continuous, garment-integrated physical monitoring of breast tissue.

I) INTRODUCTION

Despite continuous advances in prevention and treatment strategies, breast cancer remains a major global public health challenge [1]. With approximately 2 million new cases diagnosed every year and an expected increase to nearly 3 million by 2040, it represents one of the leading causes of premature mortality among women [2]. To date, breast tissue assessment relies on established diagnostic techniques, including mammography, ultrasound, magnetic resonance imaging, and elastography, which are fundamental for lesion detection, characterization, and follow-up [3], [4], [5]. Although these techniques represent the clinical gold standard, they require dedicated equipment, specialized personnel, and clinical settings, making them less suitable for frequent and preventive monitoring [6].

This limitation has encouraged growing interest in complementary strategies based on the detection of local physical parameters through non-invasive technologies integrated into conformable devices [7]. This interest is supported by the fact that breast lesions can locally alter tissue properties compared with the surrounding healthy region [8], [9]. From a mechanical perspective, tumors or nodules may increase local stiffness; from a thermal perspective, changes in metabolic activity, perfusion, and vascularization may induce superficial or subsurface temperature variations [10], [11]. Although these parameters alone are not sufficient for diagnosis, their assessment may provide complementary information on the local state of the tissue.

In this context, clinical palpation remains a simple and non-invasive procedure widely used for preliminary breast assessment [12], [13]. It exploits the different mechanical consistency between healthy tissue and altered regions, allowing clinicians to manually perceive areas characterized by increased stiffness or irregularity [14]. However, palpation is qualitative and strongly dependent on the operator's experience, the applied force, lesion depth and size, and tissue anatomy [15], [16]. Moreover, due to the limited thermal sensitivity of human touch, local temperature variations cannot be reliably detected or quantified through manual examination alone [17], [18].

These considerations have stimulated the development of wearable technologies capable of acquiring local physical information as a complement to conventional diagnostic methods [6], [7]. Devices

integrated into everyday garments could enable the monitoring of mechanical and thermal parameters under more natural conditions and over prolonged periods [19], [20], [21], [22]. However, many current solutions still show limitations in terms of spatial resolution, distributed sensing capability, and multiparametric monitoring. Therefore, there remains a need for wearable platforms combining conformability, garment integration, and distributed acquisition of complementary physical information, such as strain and temperature.

In this scenario, optical fiber sensors represent a particularly promising solution. Their small size, lightweight structure, flexibility, immunity to electromagnetic interference, and compatibility with soft polymeric matrices make them well suited for integration into thin and conformable biomedical devices [23]. Moreover, the ability of specific sensing solutions based on fiber optic technology to detect both strain and temperature is consistent with applications aimed at monitoring tissue thermomechanical alterations. In biomedical applications, several optical fiber sensing approaches have been investigated, with a large number of devices relying on point or discrete sensing elements, such as fiber Bragg gratings and Fabry–Perot interferometric sensors [24], [25]. They have been employed for physiological and biomechanical monitoring, biosensing, tactile probes, medical tools, and devices for minimally invasive surgery, aimed at detecting local variations in force, strain, or stiffness associated with tissue alterations [26], [27], [28], [29], [30], [31], [32], [33], [34], [35]. However, point-based optical fiber sensors provide measurements only at the specific locations where the sensing elements are implemented. Even in multipoint or array-based configurations, such as Bragg grating arrays, spatial coverage and resolution depend on the number and position of the individual sensing elements, often requiring prior knowledge of the regions of interest [36].

A strategy to overcome these limitations is represented by distributed optical fiber sensing (DOFS), in which the entire fiber acts as a distributed sensing element [37]. Techniques based on Rayleigh backscattering and optical frequency-domain interrogation enable spatially resolved strain and temperature profiles to be acquired along the whole fiber, with a high density of measurement sites and, in some cases, sub-millimetric spatial resolution [38]. In this way, a single fiber can act as a continuous network of sensing sites, detecting strain, temperature, and derived quantities such as force, without complex arrays of point sensors. The integration of the fiber into planar geometries and soft matrices makes DOFS particularly attractive for thin, conformable wearable platforms compatible with everyday garments [39].

This study proposes a proof of concept of a soft DOFS-based wearable device (DWD), designed for future integration into everyday garments. The device consists of a thin, conformable annular silicone patch embedding a single optical fiber interrogated to perform DOFS. Its sensing layout was bioinspired by the capture spiral of orb-weaving spider webs, where a continuous silk filament is arranged over an extended surface with regular spacing and efficient area coverage. Beyond geometry, spider webs can also be regarded as distributed mechanosensory structures: local perturbations occurring on the web are transmitted through the silk network as mechanical signals, providing information on the presence and location of the stimulus [40], [41]. This functional concept, in which an extended filamentous structure acts as both a mechanical transduction medium and a spatial localization pathway, motivated the translation toward a DOFS-based layout. In the proposed DWD, this principle was implemented through an Archimedean fiber path, selected as an engineering simplification able to provide regular spacing, continuous coverage of the annular sensing region, and straightforward spatial remapping of the fiber coordinates. In this configuration, the embedded optical

fiber acts as a continuous distributed sensing element, converting local strain- and temperature-related variations into spatially resolved spectral-shift changes along the sensing path. In the envisioned application scenario, the DWD would be incorporated into a daily-use wearable support and positioned in contact with the breast surface. The gentle compression exerted by the garment would generate a distributed palpation-like mechanical interaction between the patch and the underlying tissue. In the presence of a localized inhomogeneity, such as a stiffer or warmer region, this interaction could produce local variations in the strain and temperature fields transferred to the embedded fiber. Through DOFS interrogation, these variations would be detected as spatially resolved spectral-shift changes along the fiber and remapped over the sensing surface of the patch. This study presents the bioinspired design, fabrication, metrological characterization, and preliminary phantom-based assessment of the proposed DWD. The obtained results support the feasibility of the proposed approach and highlight the potential of the DWD as a complementary wearable platform for distributed physical monitoring of breast tissue.

II) DESIGN, FABRICATION AND WORKING PRINCIPLE OF THE WEARABLE DEVICE

2.1 Design Rationale and Bioinspired Optical Fiber Layout

The proposed DWD was conceived to monitor an approximately circular breast surface region through a thin and conformable annular patch. As shown in Fig. 1(A), the device was designed with an outer diameter of 130 mm, a central opening of 35 mm, and an overall thickness of 3 mm. The annular geometry was selected to match the target anatomical region while leaving the areola–nipple area free from direct contact [42]. At the same time, the circular outer shape provides an extended and symmetric surface suitable for hosting the sensing layout and promoting conformal contact with the breast surface.

The sensing element consists of a single continuous optical fiber embedded within the silicone matrix. Since, in DOFS, the entire fiber axis contributes to the measurement, the spatial distribution of the fiber directly determines the sensing coverage of the device. For this reason, the fiber layout was bioinspired by the capture spiral of orb-weaving spider webs, and in particular by the architecture of *Araneus diadematus* [43]. In these natural structures, a continuous silk filament is distributed over a planar area with nearly regular spacing between consecutive turns, providing efficient surface coverage and mechanical signal transmission [44]. From a geometrical perspective, the capture spiral was translated into an Archimedean trajectory to cover an extended annular surface with a single continuous optical fiber. In this configuration, the optical fiber replaces the silk filament, while the capture spiral is reproduced as a controlled sensing path, extending the intrinsically one-dimensional sensing capability of the fiber over an approximately two-dimensional region.

In polar coordinates, the resulting fiber trajectory can be described as:

$$r(\theta) = R_{out} - \left(\frac{p}{2\pi}\right) \cdot \theta \quad (1)$$

where $r(\theta)$ is the radial coordinate, which decreases from the external radius $R_{out} = 57 \text{ mm}$ to the internal radius $R_{in} = 20 \text{ mm}$ as the spiral develops inward. The parameter $p = 7.4 \text{ mm}$ represents the pitch between consecutive turns.

This geometrical layout offers three properties directly relevant to distributed fiber sensing. First, as a single non-self-intersecting continuous path, it avoids fiber crossings or junctions, which would otherwise compromise mechanical robustness and signal continuity [45]. Second, the constant pitch p ensures a uniform spatial sampling density across the sensing surface, consistent with the regular turn spacing observed in the biological structure [40]. Third, the smooth and continuously varying curvature of the Archimedean spiral avoids the sharp bends associated with grid or serpentine layouts, which is particularly relevant given the bend-sensitivity of optical fibers to signal loss and localized strain artifacts.

Beyond this geometrical analogy, the layout also reflects a functional principle observed in orb-web architectures. In natural webs, radial, frame, and spiral threads form a mechanically continuous network through which prey-induced perturbations propagate as vibrations [41]. The geometry of this network, together with prestress and non-uniform tension distribution, shapes vibration transmission, transverse stiffness, modal response, and the web-mediated localization process [40]. Similarly, in the DWD, the continuous optical fiber acts as a spatially addressable path in which local strain- and temperature-related perturbations are encoded and subsequently remapped over the sensing surface. Thus, the embedded fiber provides both distributed sensing and the spatial framework used to localize perturbations within the patch, as assessed in Sections III and IV.

A schematic representation of the bioinspired optical fiber layout and its main geometrical parameters functional characteristics is reported in Fig. 1(B). Accordingly, in all plots reporting the response of the DWD as a function of position along the spiral, 0 mm corresponds to the outer end of the fiber path, while the final position corresponds to the inner end of the spiral.

2.2 Materials and Fabrication

A standard single-mode optical fiber was used as the distributed sensing element, while Ecoflex™ 00-30 two-component silicone rubber (Smooth-On Inc., USA) was selected as the soft matrix material due to its high deformability, resilience, low elastic modulus, and suitability for soft wearable devices [46]. The fiber was positioned approximately at the mid-plane of the 3 mm silicone thickness to provide mechanical protection while preserving device flexibility and promoting strain and temperature transfer from the external surface to the sensing region.

The fabrication process was based on two consecutive silicone casting steps, designed to integrate the optical fiber directly during device fabrication. Two PLA molds, produced by fused deposition modeling, were used for this purpose. The first mold was used to fabricate the first silicone layer and to reproduce the Archimedean spiral as a groove, defining the intended fiber path. After curing, this first silicone layer was removed, flipped, and positioned inside the second mold. Once the first layer was placed in the second mold, the optical fiber was manually inserted into the spiral groove, avoiding sharp curvatures and reinforcing the inlet and outlet regions with protective sheaths. A second silicone casting was then performed to fully embed and stabilize the fiber within the elastomeric matrix.

A schematic representation of the fabrication process and of the main assembly steps is reported in Fig. 1(C).

2.3 Working Principle

DOFS enables spatially resolved measurements along an optical fiber by exploiting the Rayleigh backscattered light generated during propagation [47]. Rayleigh backscattering originates from microscopic refractive-index inhomogeneities along the fiber core, which provide a characteristic spectral signature of the fiber [48]. When the fiber is subjected to axial strain or temperature variations, the local optical path length and refractive index change, producing a shift of this spectral signature with respect to a reference state.

When DOFS is performed through optical frequency domain reflectometry (OFDR) systems, the Rayleigh signature measured along the fiber is compared with a reference signature within consecutive sensing windows [49]. The shift required to maximize the correlation between the two signatures is extracted as the local spectral shift, $\Delta f(z)$, providing a distributed map of the optical response. In this framework, each sensing window can be associated with a distributed measurement site, and the distance between adjacent sites, commonly referred to as the gage pitch, defines the spatial sampling interval of the fiber response. In linearized form, the spectral shift depends on both strain and temperature according to:

$$\Delta f(z) = S_\varepsilon \Delta \varepsilon(z) + S_T \Delta T(z) \quad (2)$$

where $\Delta \varepsilon(z)$ is the strain variation, $\Delta T(z)$ is the temperature variation, S_ε and S_T are the strain and temperature sensitivity coefficients, respectively [50].

2.4 Optical Interrogation Configuration

After fabrication, the DWD was connected to the Luna ODiSI 6102 OFDR interrogator (see Fig. 1(D)) [51]. For the experimental tests, the system was configured with a gage pitch of 2.6 mm and an acquisition frequency of 80 Hz. The gage pitch defines the distance between adjacent distributed sensing sites and therefore determines the spatial sampling of the fiber response. Considering an embedded fiber length of approximately 1.2 m, this configuration provided 458 distributed measurement sites along the sensing path, enabling the spatial and temporal analysis of the thermomechanical response of the DWD.

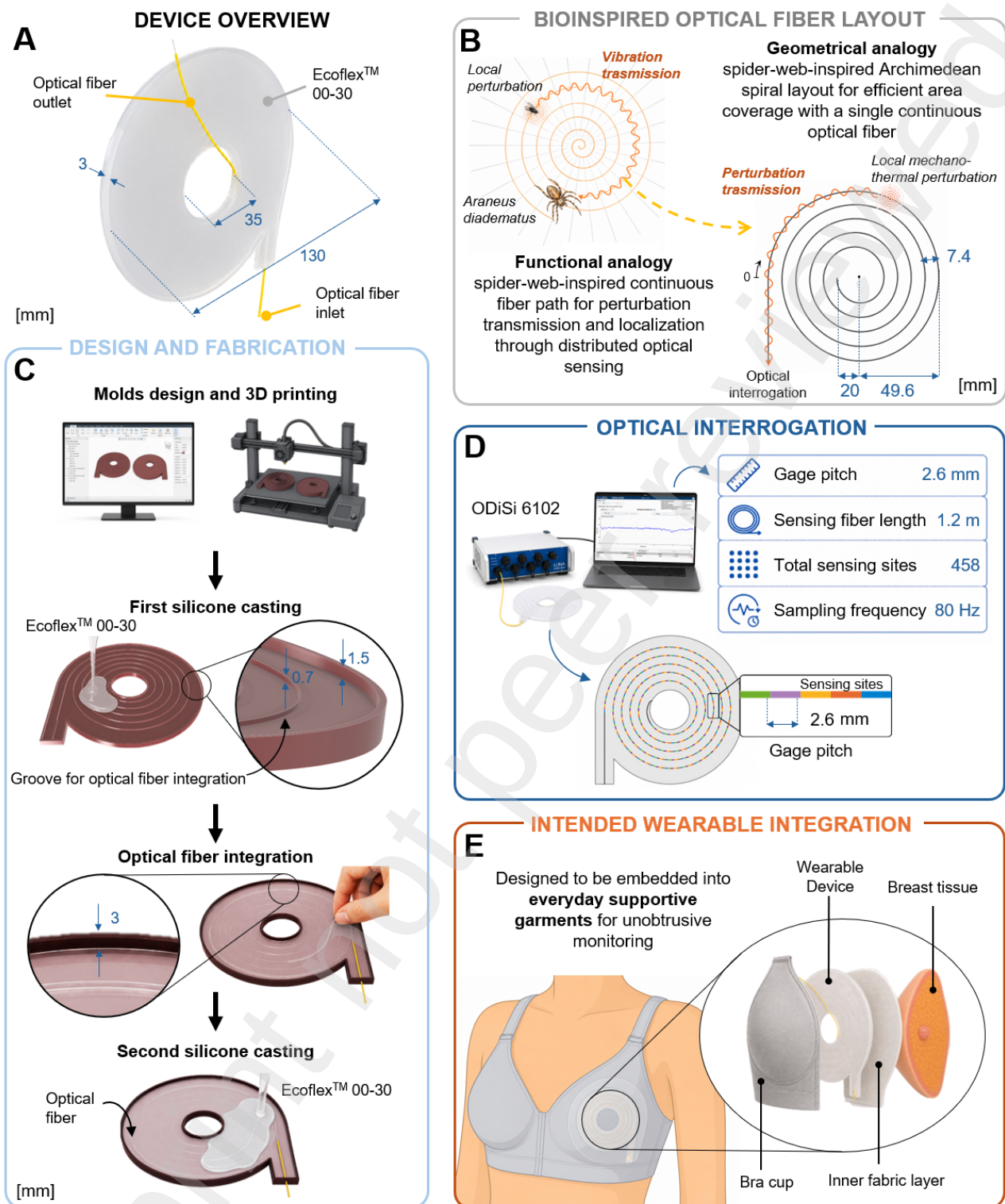


Figure 1. (A) Overall device geometry, showing the annular silicone patch, the central opening, the optical fiber inlet and outlet, and the main geometrical dimensions. (B) Bioinspired optical fiber layout together with the main layout parameters. (C) Design and fabrication workflow, including CAD modeling, mold fabrication, two-step silicone casting, optical fiber positioning within the spiral groove, and final encapsulation of the sensing path. (D) Optical interrogation configuration showing the distributed sensing sites generated along the embedded fiber and the main acquisition parameters. (E) Intended wearable integration of the device into an everyday garment for future breast tissue monitoring applications.

III) METROLOGICAL CHARACTERIZATION

Since the DWD was conceived for future integration into a wearable support gently compressed against the breast surface, as schematically shown in Fig. 1(E), it was characterized with respect to the two physical quantities most relevant to the intended application: force and temperature. Mechanical characterization was performed under force-controlled conditions, using the applied load as a controlled input to assess the distributed response of the device to compression-related perturbations. Thermal characterization was instead conducted to evaluate the influence of temperature variations on the acquired signal. Accordingly, this section describes the experimental setups, measurement protocols, data processing procedures, and main results of both characterization campaigns.

3.1 Static Force Characterization

The static force characterization was performed using a universal testing machine (model 3565, Instron Inc., USA) to apply quasi-static compressive loads over the entire patch surface. As shown in Fig. 2(A), the device was tested in a multilayer configuration representative of the intended wearable application. The DWD was placed between a cylindrical EcoflexTM GEL 2 (Smooth-On Inc., USA) phantom, used to mimic the mechanical behavior of soft biological tissue, and a cylindrical DragonSkinTM 20 layer (Smooth-On Inc., USA), used to reproduce the compliant interface of an everyday garment [52]. The lower support and the upper loading element were fabricated in PLA by 3D printing to ensure stable positioning and a uniform load distribution over the sensing area.

Ten force-controlled compression cycles up to 100 N were performed at a displacement rate of 1 mm/min to ensure quasi-static conditions. The maximum load was selected to provide a reproducible, clinically inspired global compression condition. Mammographic breast compression was considered as an order-of-magnitude reference, since it represents a clinical scenario in which the breast is globally compressed and the resulting pressure distribution has been experimentally investigated [53].

During each cycle, the applied load, measured by a 500 N load cell mounted on the actuator, and the distributed optical response of the DWD, acquired through the ODiSI 6102 interrogator, were simultaneously recorded at 80 Hz. The temporal and spatial evolution of the distributed response is reported in Fig. 2(B,C), showing both the spectral shift acquired along the spiral during the repeated compression cycles and the spatial response maps obtained at selected force levels.

For the quantitative analysis, force–spectral shift calibration curves were obtained for each sensing site by averaging the synchronized responses over the ten compression cycles. Linear regression was used to estimate the local force sensitivity, while linearity error, uncertainty, and sensitivity distribution were evaluated to assess the metrological behavior of the device. Details on signal synchronization, uncertainty estimation, linearity error calculation, and statistical descriptors are reported in the Supplementary Material S1.

The global force–spectral shift response is shown in Fig. 2(D), while representative calibration curves and the spatial distribution of force sensitivity are reported in Fig. 2(E,F,G). In all cases, increasing compressive force produced a progressive decrease in the spectral shift, indicating a monotonic response with negative sensitivity. The local calibration curves showed a nearly linear behavior, with a median R^2 of 0.99 and a mean linearity error of 3.02%, ranging from approximately 1% to 7%. Over

all valid sensing sites, the force sensitivity ranged from -1.12 to -0.12 GHz/N, with a mean value of -0.71 GHz/N.

The sensitivity distribution highlighted a clear spatial dependence along the fiber. In particular, the magnitude of the force sensitivity was generally higher in the outer portions of the spiral and progressively decreased toward the inner regions. The sensitivity coefficient of variation of 36.00% reflects this spatial variability and may also be related to non-uniform contact and load transfer at the phantom–DWD interface during compression. This behavior is consistent with the distributed nature of the device and confirms the importance of considering the local response of each sensing site when interpreting spatial maps. The main global metrological parameters obtained from the static force characterization are summarized in Table I.

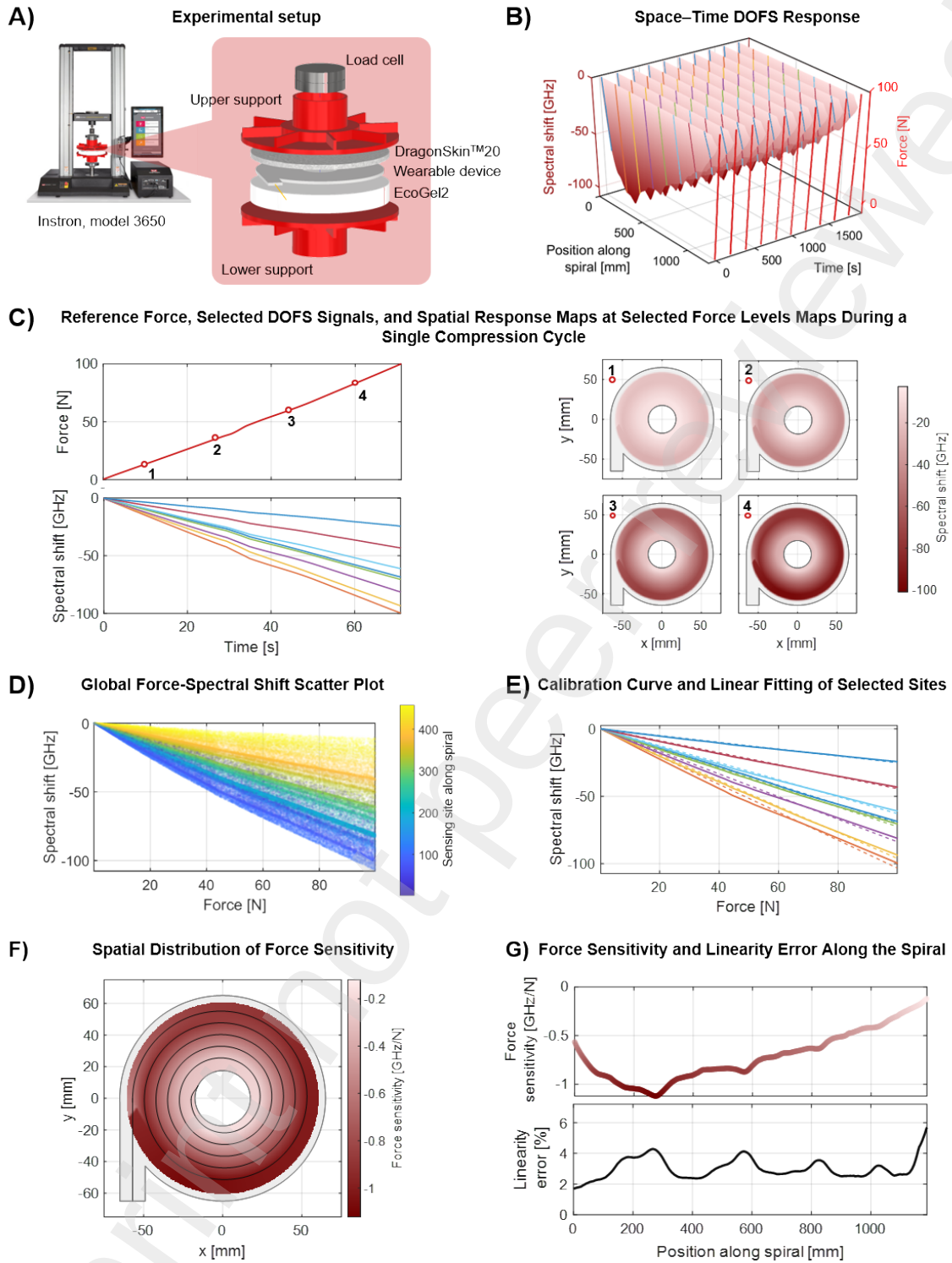


Figure 2. Static force metrological characterization of the DOFS-based wearable device. (A) Experimental setup used for the compression tests. (B) Space-time representation of the distributed spectral shift acquired along the spiral during the ten compression repetitions, together with the applied force signal. (C) Reference force, selected DOFS signals, and spatial response maps at selected force levels during a single compression cycle. (D) Global force-spectral shift scatter plot including all valid measurement sites distributed along the spiral. (E) Calibration curves of selected representative sites, together with the corresponding linear regression lines and uncertainty bands. (F) Spatial distribution of the force sensitivity remapped over the sensing surface of the patch. (G) Force sensitivity profile along the spiral and corresponding linearity error profile.

Table I. Global metrological parameters from static force characterization.

Metrological Parameter	Value
Mean uncertainty	0.15 GHz
Minimum uncertainty	0.05 GHz
Maximum uncertainty	0.27 GHz
Median R^2	0.99
Mean linearity error	3.02 %
Minimum linearity error	1.40 %
Maximum linearity error	6.67 %
Mean force sensitivity	-0.71 GHz/N
Minimum force sensitivity magnitude	0.12 GHz/N
Maximum force sensitivity magnitude	1.12 GHz/N
Sensitivity coefficient of variation	36.00 %

3.2 Static Temperature Characterization

The static temperature characterization was performed using a climatic chamber (KBF-S ECO E6, Binder, Germany) to expose the DWD to controlled and reproducible thermal variations. As shown in Fig. 3(A), both the sensing patch and a reference temperature sensor (EL-USB-TP-LCD, Lascar Electronics, England) were placed inside the chamber. Three consecutive heating and cooling cycles were performed over the temperature range from 22 °C to 44 °C. This interval was selected to include values close to room and body surface temperature, as well as higher thermal levels potentially relevant to localized temperature alterations associated with the intended biomedical application [11], [54].

Each complete heating and cooling cycle lasted approximately 3 h, imposing slow temperature variations and maintaining the system under quasi-static conditions. During the entire test, the distributed optical response of the DWD and the reference temperature signal were simultaneously acquired at 1 Hz. The temporal and spatial evolution of the distributed thermal response is reported in Fig. 3(B,C), showing both the spectral shift acquired along the spiral during the three thermal cycles and the spatial response maps obtained at selected temperature levels.

For the quantitative analysis, only the heating phases of the three cycles were considered. Temperature variation–spectral shift calibration curves were obtained for each sensing site by averaging the synchronized responses over the three heating phases. Linear regression was used to estimate the local thermal sensitivity, while linearity error, uncertainty, and sensitivity distribution were evaluated to assess the thermal metrological behavior of the device. Details on signal segmentation, synchronization, uncertainty estimation, linearity error calculation, and statistical descriptors are reported in the Supplementary Material S1.

The global temperature variation–spectral shift response is shown in Fig. 3(D), while representative calibration curves and the spatial distribution of thermal sensitivity are reported in Fig. 3(E, F, G). In all cases, increasing temperature produced a progressive decrease in the spectral shift, indicating a monotonic response with negative thermal sensitivity. The local calibration curves showed a strongly linear behavior, with a median R^2 of 0.99 and a mean linearity error of 1.71%, ranging from

approximately 1% to 5%. Over all valid sensing sites, the thermal sensitivity ranged from -2.70 to -1.79 GHz/°C, with a mean value of -2.25 GHz/°C.

The thermal sensitivity distribution showed a spatial dependence along the fiber. In particular, the magnitude of the thermal sensitivity was generally higher in the initial and outer portions of the spiral and progressively decreased toward the inner regions. However, the coefficient of variation was 11.37%, markedly lower than that obtained for force sensitivity, indicating a more uniform thermal response. This difference may be attributed to the lower influence of non-uniform mechanical contact and load transfer at the phantom–patch interface during thermal stimulation. The main global metrological parameters obtained from the static temperature characterization are summarized in Table II.

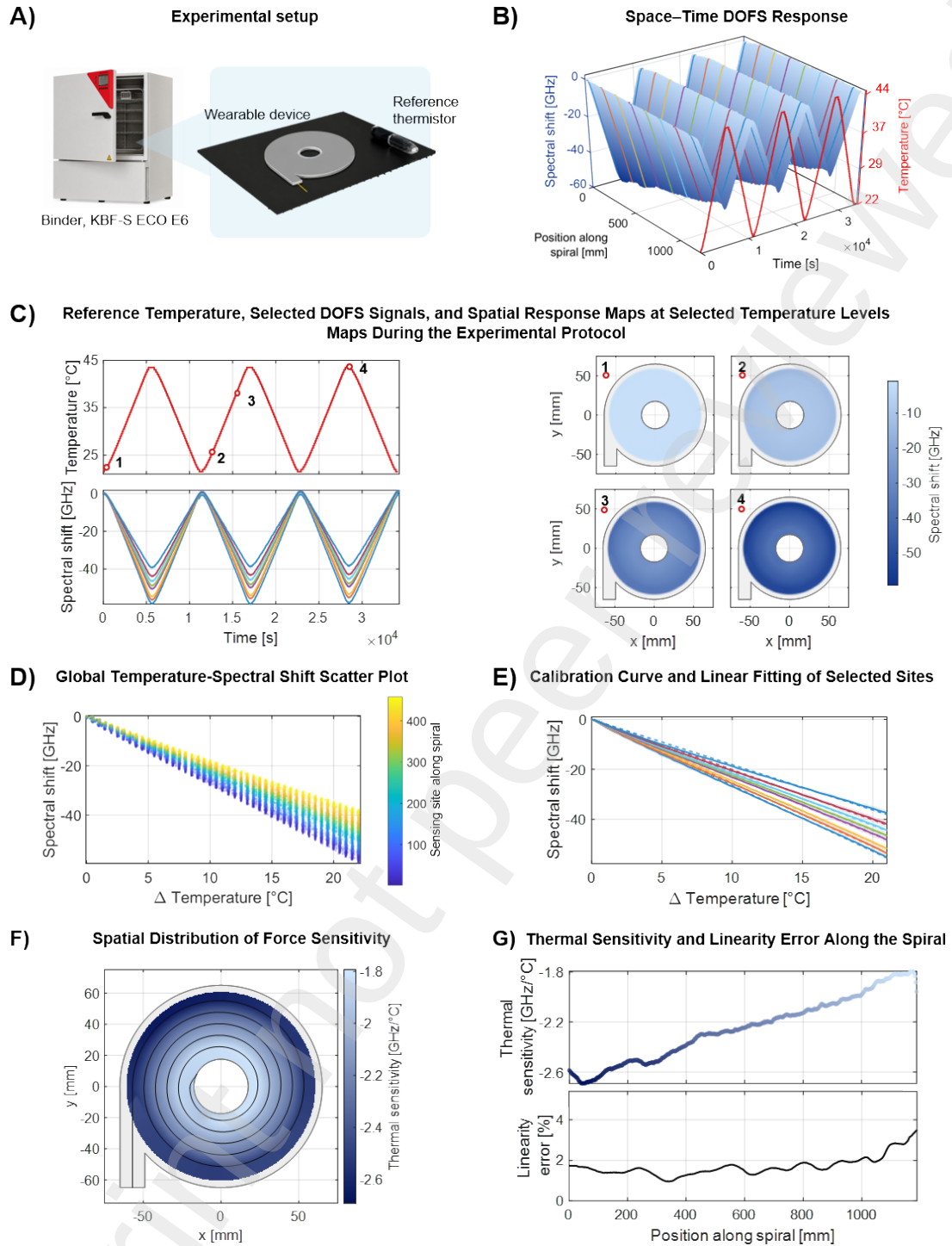


Figure 3. Static temperature metrological characterization of the DOFS-based wearable device. (A) Experimental setup used for the thermal tests. (B) Space-time representation of the distributed spectral shift acquired along the spiral during the three thermal cycles, together with the reference temperature signal. (C) Reference temperature, selected DOFS signals, and spatial response maps at selected temperature levels during a representative thermal cycle. (D) Global temperature variation-spectral shift scatter plot including all valid measurement sites distributed along the spiral. (E) Calibration curves of selected representative sites, together with the corresponding linear regression lines and uncertainty bands. (F) Spatial distribution of the thermal sensitivity remapped over the sensing surface of the patch. (G) Thermal sensitivity profile along the spiral and corresponding linearity error profile.

Table II. Global metrological parameters from static temperature characterization.

Metrological Parameter	Value
Mean uncertainty	0.68 GHz
Minimum uncertainty	0.51 GHz
Maximum uncertainty	1.01 GHz
Median R^2	0.99
Mean linearity error	1.71 %
Minimum linearity error	0.81 %
Maximum linearity error	4.80 %
Mean temperature sensitivity	-2.25 GHz/°C
Minimum temperature sensitivity magnitude	1.79 GHz/°C
Maximum temperature sensitivity magnitude	2.70 GHz/°C
Sensitivity coefficient of variation	11.37 %

IV) PHANTOM-BASED ASSESSMENT OF LOCAL THERMOMECHANICAL ALTERATIONS

Following the static metrological characterization, preliminary phantom-based tests were performed to assess the ability of the DWD to detect and spatially localize simulated thermomechanical alterations under controlled conditions. The metrological results obtained from the previous characterization were used as the basis for the data analysis and interpretation of the phantom experiments. Two separate validation tests were carried out, targeting localized mechanical and thermal perturbations, respectively. The following sections describe the experimental setup, protocol, data analysis, and main results of each test.

4.1 Detection of Simulated Mechanical Inhomogeneities

The first phantom-based test was designed to evaluate the ability of the DWD to detect localized mechanical inhomogeneities embedded within a soft medium. To this aim, a disc-shaped Ecoflex™ GEL 2 phantom was fabricated to mimic the geometry and compliance of the breast region. During fabrication, a rigid PLA inclusion with a diameter of 15 mm was embedded inside the phantom at a distance of 2 mm from its upper surface, providing a simplified model of a localized stiffness contrast. Mechanical tests were performed using the same compression setup adopted for the static force characterization. In this case, the phantom containing the rigid inclusion was placed below the DWD, and a compressive load from 0 to 100 N was applied over the whole multilayer system. Fig. 4(A) shows the experimental setup and the phantom with the inclusion employed for the test. Four configurations were tested by changing the relative position of the inclusion with respect to the DWD, corresponding to the north, east, south and west regions of the phantom.

The DOFS data were processed using the local force sensitivities obtained from the previous static characterization. First, the DWD spectral shift signals were synchronized with the load-cell signal, so that each distributed optical profile could be associated with the corresponding value of the applied force. For each sensing site, the spectral shift was converted into an equivalent-force response as:

$$F_{DWD}(t,i) = \frac{\Delta f(t,i)}{S_F(i)} \quad (3)$$

where $\Delta f(t,i)$ is the spectral shift measured at the i -th sensing site at the selected time instant t , corresponding to a given force level, and $S_F(i)$ is the local force sensitivity of the same site. Since the test was performed under force-controlled conditions, deviations from the expected uniform response were evaluated by subtracting the applied force measured by the load cell at the same time instant:

$$\Delta F(t,i) = F_{DWD}(t,i) - F_{app}(t) \quad (4)$$

where $F_{app}(t)$ corresponds to the applied force recorded by the load cell at the t instant of time. This operation provided a differential equivalent-force matrix, in which the rows corresponded to the synchronized time instants and the columns to the sensing sites distributed along the fiber.

To emphasize localized deviations and reduce slowly varying spatial components, a moving-average filter was applied along the sensing sites of the differential matrix. The filtered spatial trend was then subtracted from $\Delta F(i,t)$, obtaining a residual matrix representative of the local deviations from the expected homogeneous response. The residual matrix was subsequently normalized between 0 and 1 by considering the global minimum and maximum residual values calculated over the entire acquisition, i.e., across all time instants and sensing sites.

Since the normalized residual matrix remained synchronized with the load-cell signal, the residual profiles corresponding to three representative force levels, namely 20 N, 50 N, and 100 N, were extracted and remapped onto the two-dimensional patch geometry according to the theoretical position of each sensing site along the Archimedean spiral. The resulting maps were therefore used to visualize the regions in which the mechanical response deviated most from the expected homogeneous condition.

The obtained maps, reported in Fig. 4(B), show the evolution of the normalized residual response for the four inclusion positions and for the three selected force levels. In all configurations, the residual distribution changed as the applied load increased, with more evident localized regions appearing at higher compression forces. The areas characterized by the highest residual values were generally consistent with the imposed position of the rigid inclusion, suggesting that the DWD was able to detect the local mechanical perturbation induced by the stiffness contrast within the phantom.

These results indicate that, under controlled phantom-based conditions, the combination of local force calibration, differential analysis, spatial detrending, and remapping can highlight localized mechanical inhomogeneities within a compliant medium.

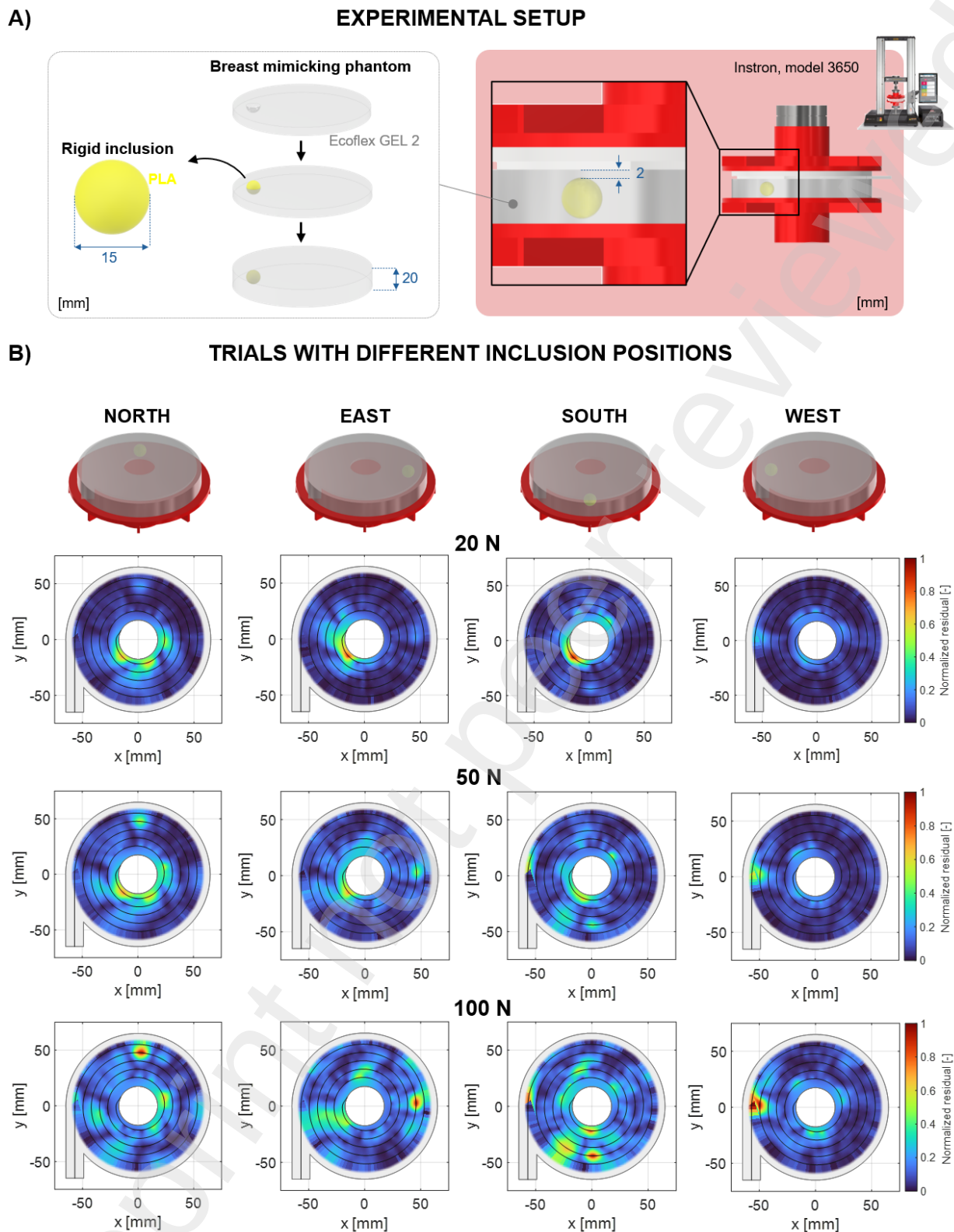


Figure 4. Phantom-based assessment of simulated mechanical inhomogeneities. (A) Experimental setup for the compression test, including the Ecoflex™ GEL 2 breast-mimicking phantom with the embedded rigid PLA inclusion and the multilayer configuration used in the Instron testing machine. (B) Normalized residual maps obtained for four inclusion positions with respect to the DWD, namely north, east, south, and west, at three applied force levels: 20 N, 50 N, and 100 N.

4.2 Detection of Simulated Thermal Inhomogeneities

The second phantom-based test was designed to evaluate the ability of the DWD to detect localized temperature increases within a soft breast-mimicking medium. This test was intended to reproduce, under controlled and simplified conditions, localized thermal perturbations that may be associated with the presence of breast lesions, without aiming to replicate the full complexity of heat generation and transfer in pathological tissue.

To this aim, a disc-shaped Ecoflex™ GEL 2 phantom was fabricated. Localized thermal perturbations were simulated by creating cavities inside the phantom, which were filled with heated water during the experimental trials. As shown in Fig. 5(A), three circular cavities were fabricated: two with the same larger diameter and one with a smaller diameter. All cavities were positioned so that their bottom surface was 2.5 mm from the flat lower surface of the phantom, corresponding to the side in contact with the DWD. In addition, a branched cavity was also produced to simulate a more irregular and spatially extended thermal pattern.

The experimental setup consisted of placing the DWD on the working plane and positioning the phantom above it, with the cavities facing upward. In this configuration, heated water could be injected into each cavity using a syringe, while the DWD remained in contact with the lower surface of the phantom. The tests were performed at an ambient temperature of approximately 25 °C. The water was heated in a beaker on a magnetic stirrer up to 60 °C before being transferred with the syringe and injected into the selected cavity. This initial reservoir temperature was set to compensate for the rapid heat loss occurring during transfer, injection, and diffusion through the phantom, so that a lower and attenuated thermal perturbation could be generated at the DWD interface.

Several trials were performed by changing the cavity in which the heated water was injected, thus varying the position of the thermal perturbation with respect to the DWD. In two representative single-cavity trials, corresponding to one large circular cavity and the smaller circular cavity, an additional FBG sensor was placed at the interface between the DWD and the phantom, at the location of the heated cavity, to provide a local temperature reference. The distributed response of the DWD was acquired using the OFDR system, while the reference FBG sensor was interrogated using a si255 optical interrogator (Luna Inc., USA). All data were then imported into MATLAB for processing.

The DWD data were processed using the local thermal sensitivities obtained from the static temperature characterization. For each sensing site, the spectral shift was converted into an estimated temperature variation as:

$$\Delta T_{DWD}(t,i) = \frac{\Delta f(t,i)}{S_T(i)} \quad (5)$$

where $\Delta f(t,i)$ is the spectral shift measured at the i -th sensing site at time t , and $S_T(i)$ is the corresponding local thermal sensitivity. The resulting distributed temperature profiles were remapped onto the two-dimensional patch geometry according to the theoretical position of each sensing site along the Archimedean spiral. This procedure allowed the temporal evolution of the localized thermal response over the DWD surface to be visualized.

The results obtained from the single-cavity trials are reported in Fig. 5(B). For each case, the map corresponding to the maximum response recorded over the DWD was selected as representative of the thermal perturbation. The maps show that the region with the highest DWD output was located

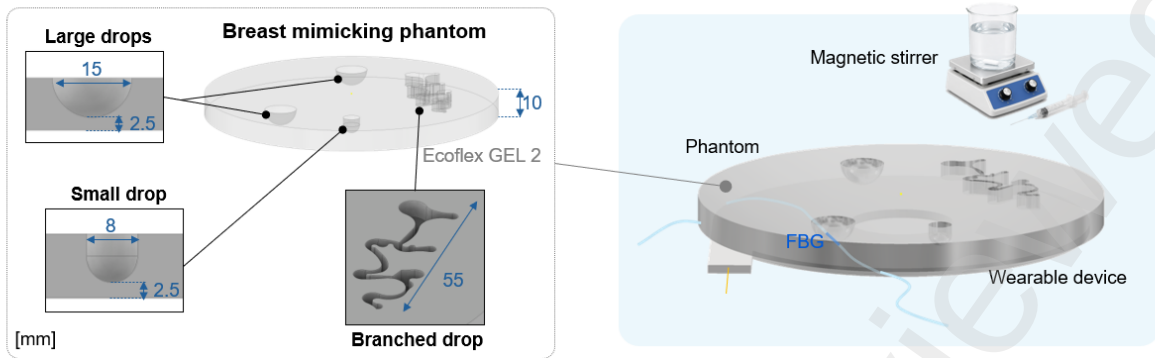
in correspondence with the heated cavity, allowing the position of the imposed thermal perturbation to be qualitatively identified. Moreover, the spatial extension of the region characterized by higher response was larger for the large cavity than for the smaller one, suggesting that the distributed response of the device was consistent with the size of the heated region.

For the same single-cavity trials, the local temporal response of the DWD was compared with the temperature variation measured by the reference FBG sensor. The Bragg wavelength shift of the reference FBG was converted into temperature variation using the known temperature sensitivity of the bare FBG [55]. In each trial, the DWD sensing site exhibiting the maximum response was selected as the central point. Four neighboring sensing sites were then identified as the sites closest to the intersections between the spiral turns and a circle centered on this point with radius equal to the spiral pitch, 7.4 mm, and their temporal responses were plotted together with the maximum-response site and the reference FBG, as shown in Fig. 5(B). Considering the ambient temperature of approximately 25 °C, the temperature variations measured by the reference FBG indicate that the local temperature reached at the phantom–DWD interface remained within the range investigated during the static temperature characterization. Moreover, as expected from heat transfer through the phantom and the soft patch, the DWD response was delayed and reduced in amplitude compared with the FBG reference signal, which was located closer to the heated region [56]. Nevertheless, the waveform of the selected DWD sensing sites followed the same overall temporal trend as the reference measurement, confirming the ability of the device to detect the local thermal perturbation.

Additional trials were performed to assess the ability of the DWD to reconstruct more complex thermal patterns. In particular, two large circular cavities were filled simultaneously to generate two localized thermal perturbations, while the branched cavity was used to impose an irregular and spatially extended heated region. The corresponding maps are shown in Fig. 5(C). In both cases, the DWD response allowed the imposed thermal pattern to be qualitatively reconstructed, highlighting either the presence of multiple heated regions or the spatial extension of the branched geometry.

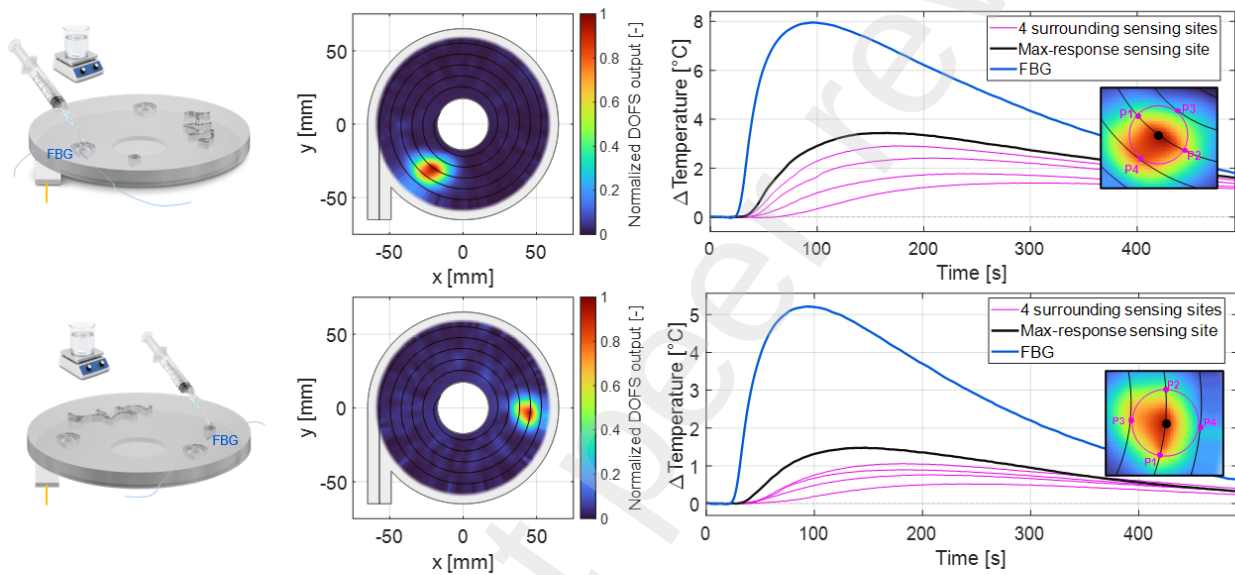
A)

EXPERIMENTAL SETUP



B)

SINGLE-CAVITY THERMAL RESPONSE WITH FBG REFERENCE



C)

RECONSTRUCTION OF MULTI-CAVITY AND BRANCHED THERMAL PATTERNS

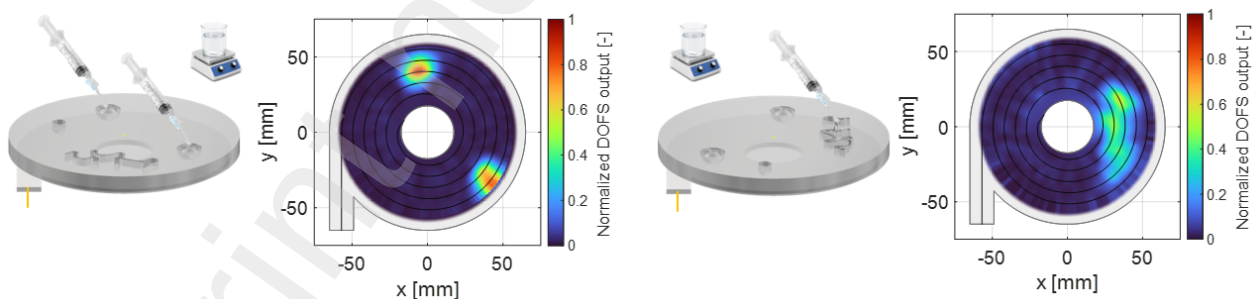


Figure 5. Phantom-based assessment of simulated thermal inhomogeneities. (A) Experimental setup, including the Ecoflex™ GEL 2 breast-mimicking phantom with circular and branched cavities, the magnetic stirrer used to heat the water, and the DWD–FBG sensing configuration. (B) Single-cavity thermal response obtained for a large and a small cavity, showing the normalized DOFS maps and the comparison between the FBG reference signal, the maximum-response DOFS site, and four surrounding sensing sites. (C) Reconstruction of more complex thermal patterns obtained by simultaneously filling two circular cavities and by filling the branched cavity.

V) DISCUSSION AND CONCLUSIONS

This work presents a proof of concept of a soft, bioinspired DOFS-based wearable device for distributed monitoring of thermomechanical alterations in breast tissue. The DWD consists of a thin annular silicone patch embedding a single continuous optical fiber arranged as an Archimedean spiral, inspired by the capture spiral of orb-weaving spider webs. Beyond considering the spider web as a geometric or structural template, this design also reflects its function as a distributed vibrational sensing and localization medium, translated here into a distributed optical sensing path capable of spatially resolving mechanical and thermal perturbations. Interrogated through OFDR, this layout provides 458 sensing sites distributed along the fiber and remapped over the annular active region, extending fiber sensing over a two-dimensional surface. The device was conceived for future integration into wearable supports, such as bras or compression bands, able to maintain contact with the breast surface and apply gentle distributed compression. In this configuration, local stiffness or temperature alterations could modify the interaction between tissue, patch, and fiber, producing spatially localized variations in the distributed response.

Static metrological characterization was essential to interpret the DWD as a distributed measuring system. Compression and temperature tests enabled the behavior of individual sensing sites along the spiral to be analyzed and the main metrological parameters to be quantified. In both cases, the DWD showed a monotonic and approximately linear response within the investigated range. The estimation of local force and temperature sensitivities also provided the calibration basis for subsequent controlled tests closer to the intended application scenario.

Phantom-based tests were then performed to assess the ability of the DWD to detect and spatially remap localized mechanical and thermal alterations. In the mechanical validation test, the setup reproduced a simplified wearable-like condition, with the patch compressed against a soft breast-mimicking phantom. A circular rigid PLA inclusion embedded in the EcoflexTM GEL 2 phantom generated higher-response regions in the normalized residual maps, generally consistent with the imposed inclusion position. Their extension and shape were qualitatively compatible with the inclusion geometry, and the perturbation became more evident as the compression force increased. Secondary regions were also observed, likely due to contact conditions, non-uniform load distribution and mechanical coupling between phantom and patch. These effects are consistent with real scenarios, where tissue compression is not perfectly uniform or strictly controlled, as also occurs during manual palpation. Nevertheless, the results suggest that local calibration, differential analysis, and spatial remapping can highlight localized mechanical perturbations within a compliant medium.

The thermal validation test further supported the potential of the device for distributed monitoring. Cavities filled with heated water were used to simulate localized temperature increases within the phantom, reproducing in a simplified way thermal alterations potentially associated with breast lesions. The obtained maps enabled the qualitative identification of heated regions, both for single circular cavities and for more complex configurations. Although the FBG reference sensor measured temperature peaks higher than those expected in realistic superficial lesion-related thermal alterations, the DOFS sites located below the heated region showed a consistent waveform, with reduced amplitude and temporal delay due to heat diffusion through the phantom and patch. Notably, the DWD response remained detectable during the lower-temperature portions of the cooling phase,

supporting its potential for sensing attenuated and localized thermal variations closer to realistic application conditions.

Overall, the results support the feasibility of the proposed bioinspired DOFS-based approach and highlight the potential of the DWD as a complementary wearable platform for distributed breast tissue monitoring. Future work will focus on realistic anatomical phantoms integrating combined mechanical and thermal alterations, repeated tests under controlled wearable-like compression, improved strategies for separating mechanical and thermal contributions, and integration into a dedicated garment prototype.

BIBLIOGRAPHY

- [1] “Breast cancer.” Accessed: Jul. 01, 2026. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/breast-cancer>
- [2] M. Arnold *et al.*, “Current and future burden of breast cancer: Global statistics for 2020 and 2040,” *The Breast*, vol. 66, pp. 15–23, Dec. 2022, doi: 10.1016/J.BREAST.2022.08.010.
- [3] S. H. Jafari *et al.*, “Breast cancer diagnosis: Imaging techniques and biochemical markers,” *J. Cell. Physiol.*, vol. 233, no. 7, pp. 5200–5213, Jul. 2018, doi: 10.1002/JCP.26379.
- [4] E. S. McDonald, A. S. Clark, J. Tchou, P. Zhang, and G. M. Freedman, “Clinical Diagnosis and Management of Breast Cancer,” *Journal of Nuclear Medicine*, vol. 57, no. Supplement 1, pp. 9S-16S, Feb. 2016, doi: 10.2967/JNUMED.115.157834.
- [5] T. B. Bevers *et al.*, “Breast Cancer Screening and Diagnosis,” *Journal of the National Comprehensive Cancer Network*, vol. 7, no. 10, pp. 1060–1096, Nov. 2009, doi: 10.6004/JNCCN.2009.0070.
- [6] A. M. Kamal, T. Sakorikar, U. M. Pal, and H. J. Pandya, “Engineering Approaches for Breast Cancer Diagnosis: A Review,” *IEEE Rev. Biomed. Eng.*, vol. 16, pp. 687–705, 2023, doi: 10.1109/RBME.2022.3181700.
- [7] A. Q. Khan *et al.*, “Advances in breast cancer diagnosis: a comprehensive review of imaging, biosensors, and emerging wearable technologies,” *Front. Oncol.*, vol. 15, p. 1587517, Jun. 2025, doi: 10.3389/FONC.2025.1587517/FULL.
- [8] M. Gautherie, “Thermopathology of breast cancer: measurement and analysis of in vivo temperature and blood flow,” *Ann. N. Y. Acad. Sci.*, vol. 335, no. 1, pp. 383–415, 1980.
- [9] A. R. Skovoroda *et al.*, “Quantitative analysis of the mechanical characteristics of pathologically changed soft biological tissues,” *Biophysics (Oxf)*, vol. 40, no. 6, pp. 1359–1364, 1995.
- [10] C. Goodbrake *et al.*, “On the three-dimensional mechanical behavior of human breast tissue,” *Ann. Biomed. Eng.*, vol. 50, no. 5, p. 601, May 2022, doi: 10.1007/S10439-022-02951-Y.
- [11] A. Lozano, J. C. Hayes, L. M. Compton, J. Azarnoosh, and F. Hassanipour, “Determining the thermal characteristics of breast cancer based on high-resolution infrared imaging, 3D breast scans, and magnetic resonance imaging,” *Scientific Reports 2020 10:1*, vol. 10, no. 1, pp. 10105–, Jun. 2020, doi: 10.1038/s41598-020-66926-6.
- [12] H. MacPherson, “Body palpation and diagnosis,” *JOURNAL OF CHINESE MEDICINE-HOVE-*, p. 5, 1994.
- [13] S. McDonald, D. Saslow, and M. H. Alciati, “Performance and Reporting of Clinical Breast Examination: A Review of the Literature,” *CA Cancer J. Clin.*, vol. 54, no. 6, pp. 345–361, Nov. 2004, doi: 10.3322.
- [14] J. G. Elmore, K. Armstrong, C. D. Lehman, and S. W. Fletcher, “Screening for Breast Cancer,” *JAMA*, vol. 293, no. 10, pp. 1245–1256, Mar. 2005, doi: 10.1001/JAMA.293.10.1245.

- [15] L. Mahoney and A. Csima, "Efficiency of palpation in clinical detection of breast cancer," *Can. Med. Assoc. J.*, vol. 127, no. 8, p. 729, 1982.
- [16] J. Konstantinova, M. Li, G. Mehra, P. Dasgupta, K. Althoefer, and T. Nanayakkara, "Behavioral characteristics of manual palpation to localize hard nodules in soft tissues," *IEEE Trans. Biomed. Eng.*, vol. 61, no. 6, pp. 1651–1659, 2014, doi: 10.1109/TBME.2013.2296877.
- [17] D. Levine, J. R. Walker, D. J. Marcellin-Little, R. Goulet, and H. Ru, "Detection of skin temperature differences using palpation by manual physical therapists and lay individuals," *Journal of Manual & Manipulative Therapy*, vol. 26, no. 2, pp. 97–101, Mar. 2018, doi: 10.1080/10669817.2018.1427908.
- [18] R. T. Murff, D. G. Armstrong, D. Lancot, L. A. Lavery, and K. A. Athanasiou, "How effective is manual palpation in detecting subtle temperature differences?," *Clin. Podiatr. Med. Surg.*, vol. 15, no. 1, pp. 151–154, 1998, doi: 10.1016/S0891-8422(23)01036-4.
- [19] L. Arcarisi, L. Di Pietro, N. Carbonaro, A. Tognetti, A. Ahluwalia, and C. De Maria, "Palpreast—A New Wearable Device for Breast Self-Examination," *Applied Sciences* 2019, Vol. 9, Page 381, vol. 9, no. 3, p. 381, Jan. 2019, doi: 10.3390/APP9030381.
- [20] S. N. Mahmood *et al.*, "A Bra Monitoring System Using a Miniaturized Wearable Ultra-Wideband MIMO Antenna for Breast Cancer Imaging," *Electronics* 2021, Vol. 10, Page 2563, vol. 10, no. 21, p. 2563, Oct. 2021, doi: 10.3390/ELECTRONICS10212563.
- [21] D. N. Elsheakh, O. M. Fahmy, M. Farouk, K. Ezzat, and A. R. Eldamak, "An Early Breast Cancer Detection by Using Wearable Flexible Sensors and Artificial Intelligent," *IEEE Access*, vol. 12, pp. 48511–48529, 2024, doi: 10.1109/ACCESS.2024.3380453.
- [22] D. N. Elsheakh, R. A. Mohamed, O. M. Fahmy, K. Ezzat, and A. R. Eldamak, "Complete Breast Cancer Detection and Monitoring System by Using Microwave Textile Based Antenna Sensors," *Biosensors* 2023, Vol. 13, Page 87, vol. 13, no. 1, p. 87, Jan. 2023, doi: 10.3390/BIOS13010087.
- [23] R. Correia, S. James, S. W. Lee, S. P. Morgan, and S. Korposh, "Biomedical application of optical fibre sensors," *Journal of Optics*, vol. 20, no. 7, p. 073003, Jun. 2018, doi: 10.1088/2040-8986/AAC68D.
- [24] M. Pulcinelli, I. Condò, V. Lavorgna, C. Massaroni, E. Schena, and D. Lo Presti, "Combining fiber Bragg grating sensors and artificial intelligence in medicine," *APL Photonics*, vol. 10, no. 3, p. 30901, Mar. 2025, doi: 10.1063/5.0251132/3338882.
- [25] D. Tosi, S. Poeggel, I. Iordachita, and E. Schena, "Fiber Optic Sensors for Biomedical Applications," *Opto-Mechanical Fiber Optic Sensors: Research, Technology, and Applications in Mechanical Sensing*, pp. 301–333, Jan. 2018, doi: 10.1016/B978-0-12-803131-5.00011-8.
- [26] D. Lo Presti *et al.*, "Fiber bragg gratings for medical applications and future challenges: A review," *IEEE Access*, vol. 8, pp. 156863–156888, 2020, doi: 10.1109/ACCESS.2020.3019138.
- [27] M. D. F. Domingues *et al.*, "Wearable Devices for Remote Physical Rehabilitation Using a Fabry-Perot Optical Fiber Sensor: Ankle Joint Kinematic," *IEEE Access*, vol. 8, pp. 109866–109875, 2020, doi: 10.1109/ACCESS.2020.3001091.
- [28] L. Li *et al.*, "Embedded FBG-Based Sensor for Joint Movement Monitoring," *IEEE Sens. J.*, vol. 21, no. 23, pp. 26793–26798, Dec. 2021, doi: 10.1109/JSEN.2021.3120995.
- [29] X. Wang, Y. Jiang, S. Xu, H. Liu, and X. Li, "Fiber Bragg Grating-Based Smart Garment for Monitoring Human Body Temperature," *Sensors* 2022, Vol. 22, Page 4252, vol. 22, no. 11, p. 4252, Jun. 2022, doi: 10.3390/S22114252.
- [30] V. Lavorgna *et al.*, "A Sensorized Forceps Combining FBG Technology and FDM Manufacturing for Enhanced Force Feedback in Minimally Invasive Surgery," *IEEE Sens. J.*, vol. 25, no. 18, pp. 34596–34605, 2025, doi: 10.1109/JSEN.2025.3594479.

- [31] C. Jia, Z. Chen, X. Lin, H. Yang, and X. Zhang, "A breathing rate sensor with diaphragm-based Fabry-Perot interferometer," *Optical Fiber Technology*, vol. 90, p. 104098, Mar. 2025, doi: 10.1016/J.YOFTE.2024.104098.
- [32] S. Pevec and D. Donlagic, "Miniature all-fiber Fabry-Perot sensor for simultaneous measurement of pressure and temperature," *Applied Optics*, Vol. 51, Issue 19, pp. 4536-4541, vol. 51, no. 19, pp. 4536-4541, Jul. 2012, doi: 10.1364/AO.51.004536.
- [33] M. Jang, J. S. Kim, K. Kang, J. Kim, and S. Yang, "Towards Finger Motion Capture System Using FBG Sensors," *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS*, vol. 2018-July, pp. 3734-3737, Oct. 2018, doi: 10.1109/EMBC.2018.8513338.
- [34] M. Pulcinelli *et al.*, "A Multi-Sensor Tactile System Based on Fiber Bragg Grating Sensors for Soft Tissue Palpation," *IEEE Sens. J.*, vol. 24, no. 16, pp. 25205-25214, 2024, doi: 10.1109/JSEN.2024.3415808.
- [35] M. Pulcinelli *et al.*, "A Multi-Sensor Tactile Imaging System for Breast Cancer Detection: Feasibility Assessment on MRI-Derived Patient-Specific Phantom," *IEEE Sens. J.*, 2026, doi: 10.1109/JSEN.2026.3655101.
- [36] D. Tosi, E. Schena, C. Molardi, and S. Korganbayev, "Fiber optic sensors for sub-centimeter spatially resolved measurements: Review and biomedical applications," *Optical Fiber Technology*, vol. 43, pp. 6-19, Jul. 2018, doi: 10.1016/J.YOFTE.2018.03.007.
- [37] Z. Katrenova, S. Alisherov, T. Abdol, and C. Molardi, "Status and future development of distributed optical fiber sensors for biomedical applications," *Sens. Biosensing Res.*, vol. 43, p. 100616, Feb. 2024, doi: 10.1016/J.SBSR.2023.100616.
- [38] Z. Ding *et al.*, "Distributed Optical Fiber Sensors Based on Optical Frequency Domain Reflectometry: A review," *Sensors 2018*, Vol. 18, Page 1072, vol. 18, no. 4, p. 1072, Apr. 2018, doi: 10.3390/S18041072.
- [39] P. W. McCarthy *et al.*, "Wearable Optical Fiber Sensors in Medical Monitoring Applications: A Review," *Sensors 2023*, Vol. 23, Page 6671, vol. 23, no. 15, p. 6671, Jul. 2023, doi: 10.3390/S23156671.
- [40] M. Lott, V. F. D. Poggetto, G. Greco, N. M. Pugno, and F. Bosia, "Prey localization in spider orb webs using modal vibration analysis," *Scientific Reports 2022 12:1*, vol. 12, no. 1, pp. 19045-, Nov. 2022, doi: 10.1038/s41598-022-22898-3.
- [41] J. Wu, T. E. Miller, A. Cicirello, and B. Mortimer, "Spider dynamics under vertical vibration and its implications for biological vibration sensing," *J. R. Soc. Interface*, vol. 20, no. 206, Sep. 2023, doi: 10.1098/RSIF.2023.0365/90501.
- [42] D. K. Avşar, A. C. Aygit, E. Benlier, H. Top, and O. Taşkınalp, "Anthropometric Breast Measurement: A Study of 385 Turkish Female Students," *Aesthet. Surg. J.*, vol. 30, no. 1, pp. 44-50, Jan. 2010, doi: 10.1177/1090820X09358078.
- [43] F. Vollrath and W. Mohren, "Spiral geometry in the garden spider's orb web," *Naturwissenschaften*, vol. 72, no. 12, pp. 666-667, 1985.
- [44] S. Zschokke, "Form and function of the orb-web," *European arachnology*, vol. 19, p. 99, 2000.
- [45] D. A. Drake, R. W. Sullivan, and J. C. Wilson, "Distributed Strain Sensing from Different Optical Fiber Configurations," *Inventions 2018*, Vol. 3, Page 67, vol. 3, no. 4, p. 67, Sep. 2018, doi: 10.3390/INVENTIONS3040067.
- [46] Z. Liao, J. Yang, M. Hossain, G. Chagnon, L. Jing, and X. Yao, "On the stress recovery behaviour of Ecoflex silicone rubbers," *Int. J. Mech. Sci.*, vol. 206, p. 106624, Sep. 2021, doi: 10.1016/J.IJMECSCI.2021.106624.
- [47] X. Bao and L. Chen, "Recent Progress in Distributed Fiber Optic Sensors," *Sensors 2012*, Vol. 12, Pages 8601-8639, vol. 12, no. 7, pp. 8601-8639, Jun. 2012, doi: 10.3390/S120708601.
- [48] L. Palmieri, L. Schenato, M. Santagiustina, and A. Galtarossa, "Rayleigh-Based Distributed Optical Fiber Sensing," *Sensors 2022*, Vol. 22, Page 6811, vol. 22, no. 18, p. 6811, Sep. 2022, doi: 10.3390/S22186811.

- [49] J. Cui, S. Zhao, D. Yang, and Z. Ding, "Investigation of the interpolation method to improve the distributed strain measurement accuracy in optical frequency domain reflectometry systems," *Applied Optics*, Vol. 57, Issue 6, pp. 1424-1431, vol. 57, no. 6, pp. 1424–1431, Feb. 2018, doi: 10.1364/AO.57.001424.
- [50] L. Schenato, M. Cappelletti, D. Orsuti, A. Galtarossa, M. Santagiustina, and S. Cola, "Long-Term Persistence of Rayleigh Signature of Optical Fibers in Harsh Environment," *Journal of Lightwave Technology*, vol. 42, no. 18, pp. 6254–6261, 2024, doi: 10.1109/JLT.2024.3414254.
- [51] "ODiSI 6000 Series | Optical Distributed Sensor Interrogator." Accessed: Jul. 27, 2025. [Online]. Available: <https://lunainc.com/product/odisi-6000-series>
- [52] V. Lavorgna, M. Pulcinelli, E. Crispi, V. Tomarchio, E. Schena, and D. Lo Presti, "An FBG-Based Soft Wearable Device for Force Monitoring During Palpation," *IEEE Sens. J.*, Apr. 2026, doi: 10.1109/JSEN.2026.3667972.
- [53] M. Dustler *et al.*, "Breast compression in mammography: Pressure distribution patterns," *Acta radiol.*, vol. 53, no. 9, pp. 973–980, Nov. 2012, doi: 10.1258/AR.2012.120238.
- [54] N. A. Espíndola, W. Pinheiro dos Santos, and R. de Cássia Fernandes de Lima, "Numerical analysis of the influence of thermophysical parameters on the surface temperature of breast substitute geometry obtained from infrared images," *International Journal of Thermal Sciences*, vol. 224, p. 110673, Jun. 2026, doi: 10.1016/J.IJTHEMALSCI.2026.110673.
- [55] V. Lavorgna *et al.*, "Dual-Material 3-D-Printed Wearable Sensors Based on FBG Technology for Physiological and Biomechanical Monitoring in Back Pain Assessment," *IEEE Sens. J.*, vol. 25, no. 10, pp. 17164–17174, 2025, doi: 10.1109/JSEN.2025.3557470.
- [56] F. P. Incropera, D. P. DeWitt, T. L. Bergman, and A. S. Lavine, *Fundamentals of heat and mass transfer*, vol. 6. Wiley New York, 1996.